

Basic Electrochemistry

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Luigi Galvani



Born: September 9, 1737, Bologna, Italy
Died: December 4, 1798 (61)
Known for: Bioelectricity (animal electricity)

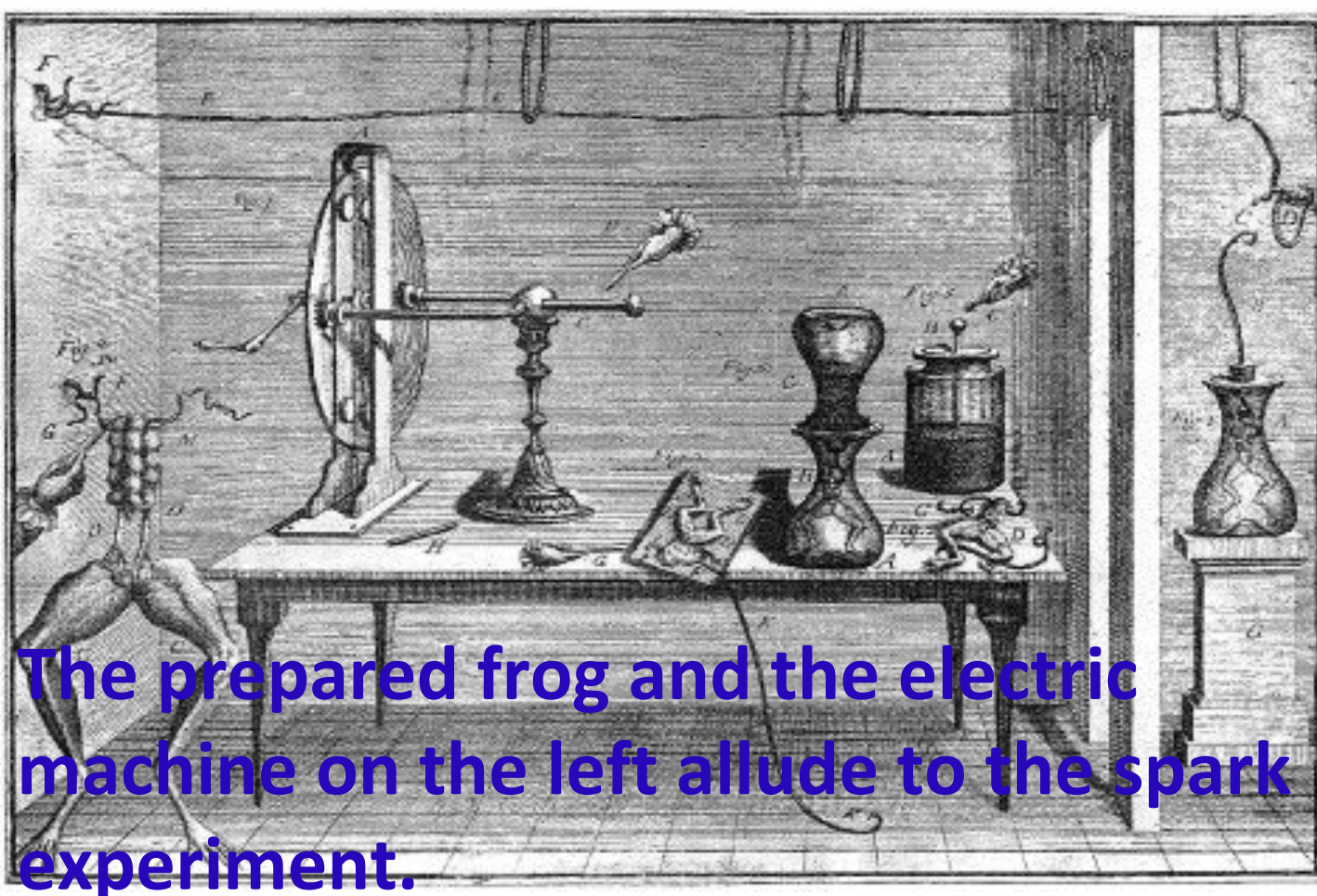
Alessandro Volta



Born: September 9, 1745, Milan, Italy
Died: March 4, 1827 (82)
Known for: Invention of electric cell, methane gas, Volt, Voltage Voltammeter

Story of Bioelectricity/ Animal electricity

- Truly It was an accidental invention
- Galvani was not a chemist or physicist rather he was a biologist and studied medicine.
- He was interested to explore his research in medical electricity

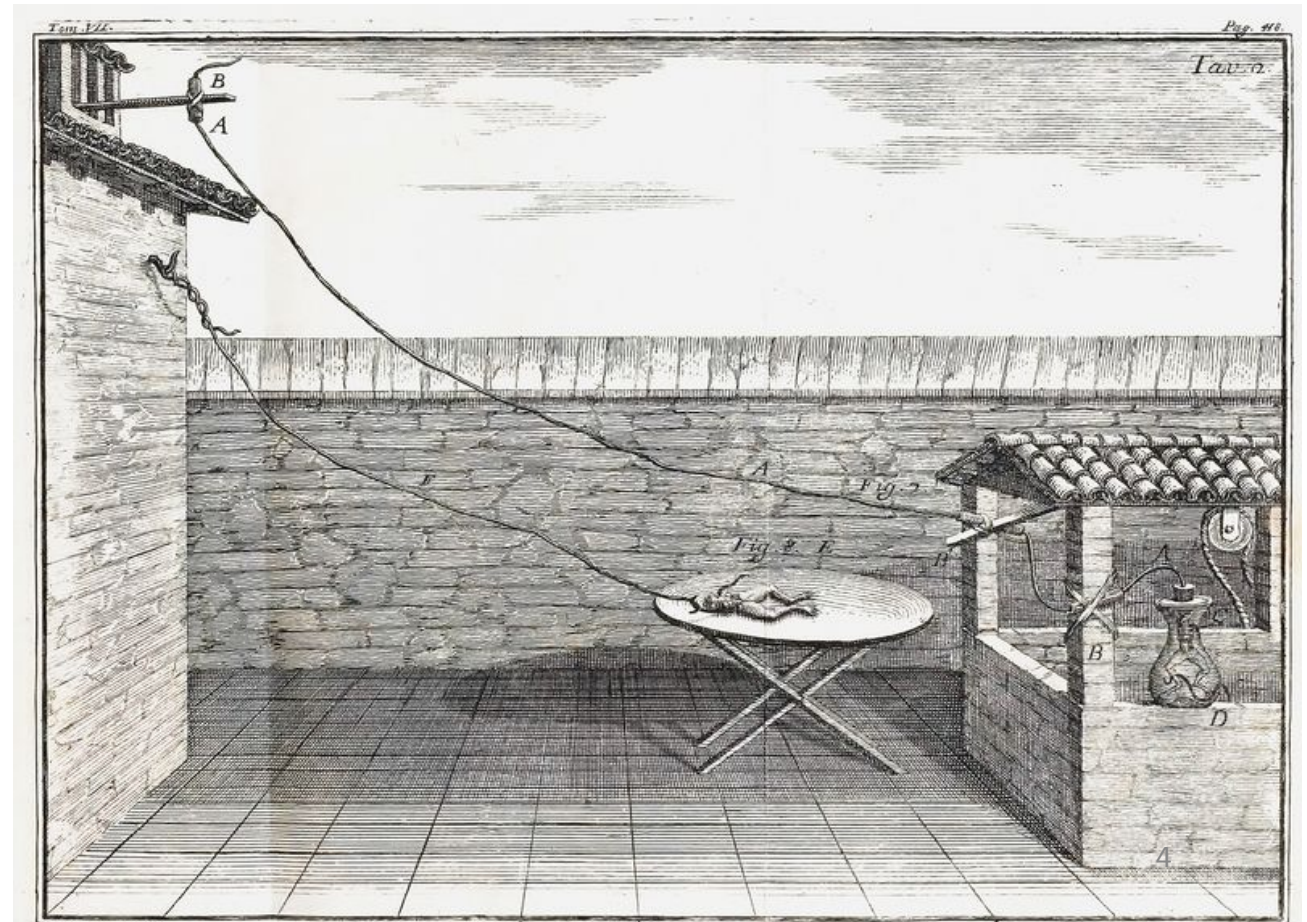


During a frog preparation, when Galvani's assistant touched an exposed sciatic nerve of the frog with a metal scalpel that had picked up a charge. At that moment, they saw sparks and the dead frog's leg kicked as if in life.

Story of Bioelectricity/ Animal electricity

To determine appropriate conditions several electric methods were used : electric machines, Leyden jars, Franklin's magic squares.

To investigate the effects of natural atmospheric electricity, one stormy evening he connected the frog nerve to a long metallic wire pointing toward the sky, obtaining strong contractions in conjunction with thunder and lightning. But in clear day nothing happened.



Story of Bioelectricity/ Animal electricity

To ascertain if quiet atmospheric electricity could also elicit contractions, on a clear day he hung prepared frogs on the iron railing of his house balcony and waited.

As nothing happened for a long time, he manipulated the frogs and, to his surprise, obtained lively contractions by pushing the metallic hooks inserted into the spinal cord towards the iron bars of the railing.

In 1786 he noticed that if the two metals he used for conduction were in contact, the muscle would still contract even when the skies were clear.



Story of Bioelectricity/ Animal electricity

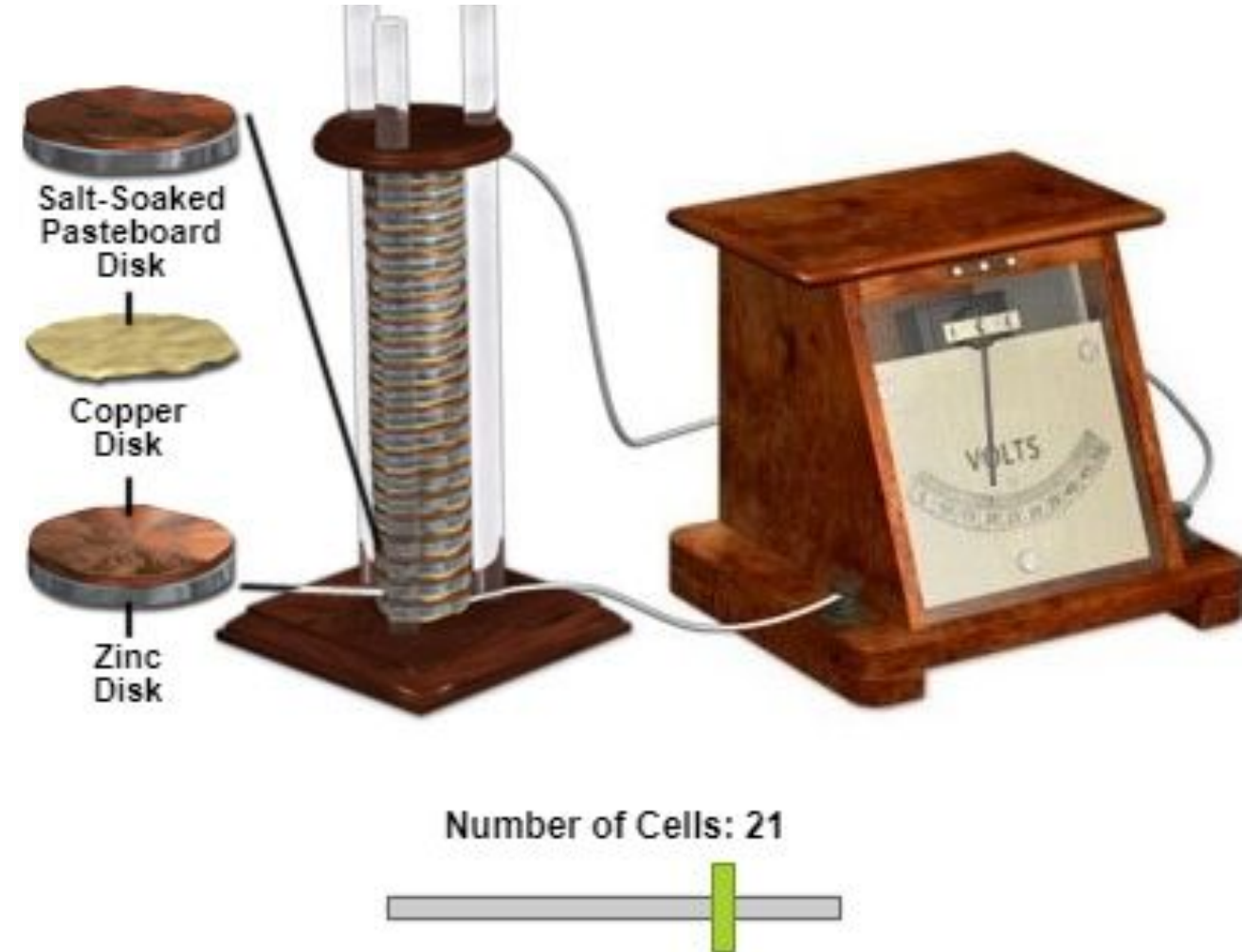
- Galvani coined the term animal electricity to describe the force that activated the muscles of his specimens.
- Along with contemporaries, he regarded their activation as being generated by an electrical fluid that is carried to the muscles by the nerves.
- The phenomenon was dubbed **galvanism**, after Galvani and his wife, on the suggestion of his peer and sometime intellectual adversary Alessandro Volta. The Galvanis are properly credited with the discovery of bioelectricity. Today, the study of galvanic effects in biology is called electrophysiology, the term galvanism being used only in historical contexts.

Volta and Galvani



Voltaic pile

- ❑ Alessandro Volta did not accept the concept of “animal electricity”.
- ❑ Rather he explained this phenomenon related to the contact between metals and moist materials.
- ❑ Volta’s work in this area led to his invention, in 1800 of the first electric battery, which came to be widely known as the voltaic pile.



Galvanic cell

Electric cells were invented about 1800 and were developed to serve practical purposes. They were not explained scientifically until about 100 years after their invention. Their use, however, contributed to scientific understanding of redox reactions and, later, this knowledge helped explain reactions inside the cell itself. Electric cells adapted for scientific study are often called galvanic cells (in recognition of Luigi Galvani) or voltaic cells (in recognition of Alessandro Volta).

galvanic cell, a device in which chemical energy is changed to electrical energy.

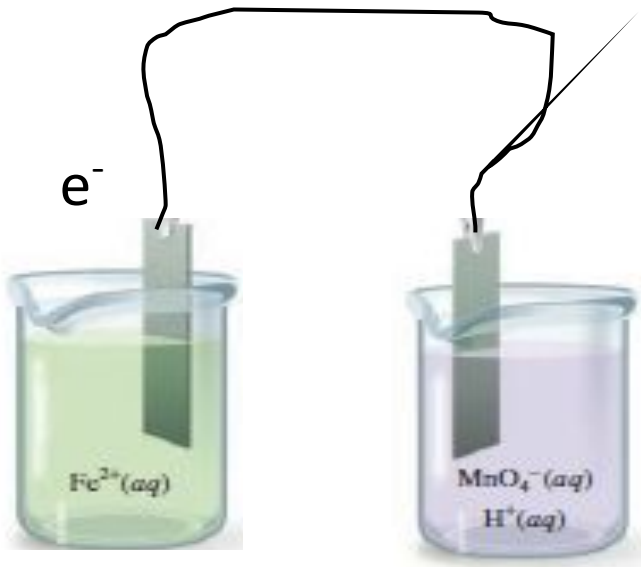
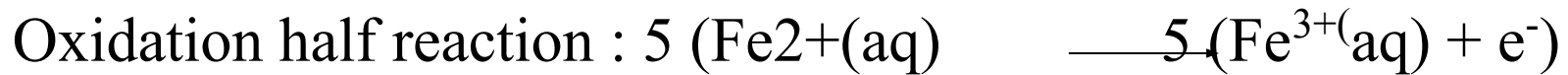
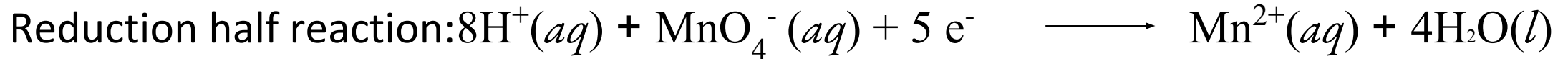
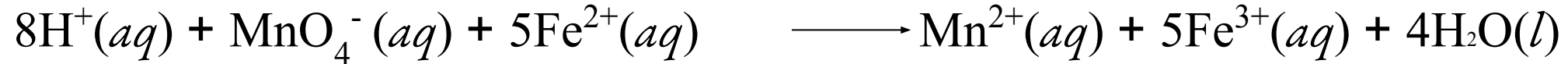
A galvanic cell uses a spontaneous redox reaction to produce a current that can be used to do work.

Oxidation occurs at the anode.

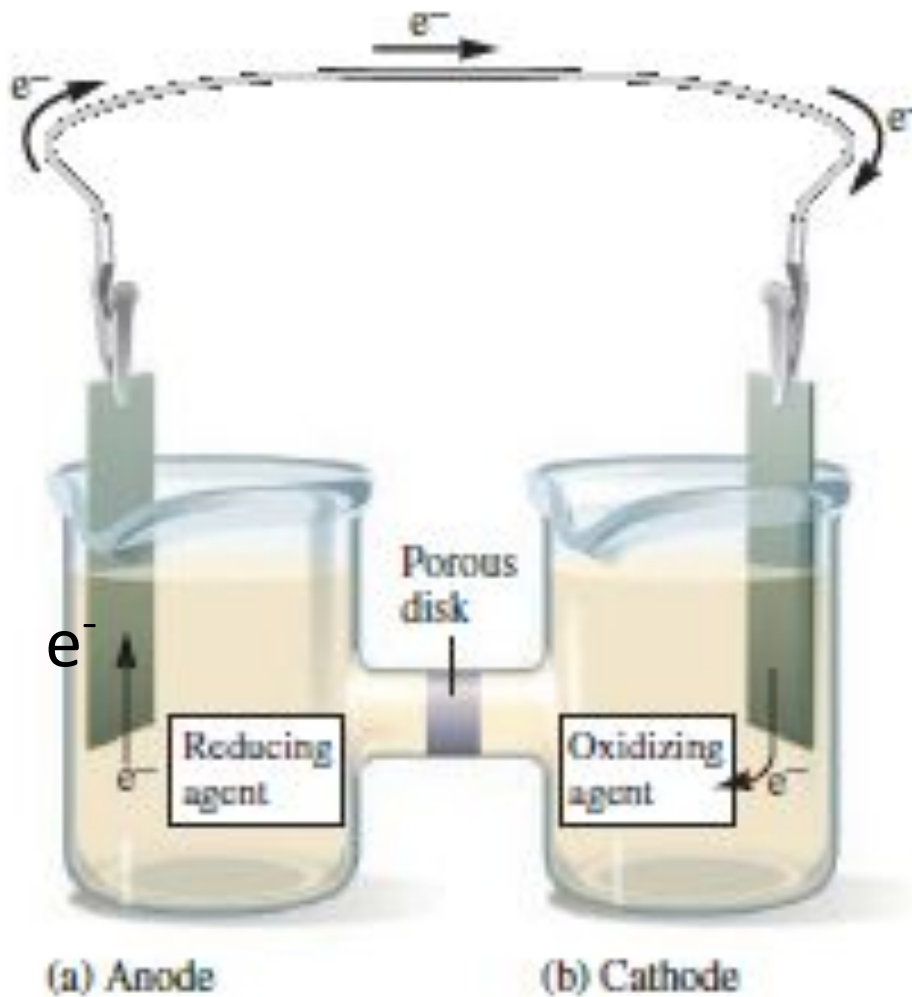
Reduction occurs at the cathode.

Galvanic cell

Let us consider the following redox reaction

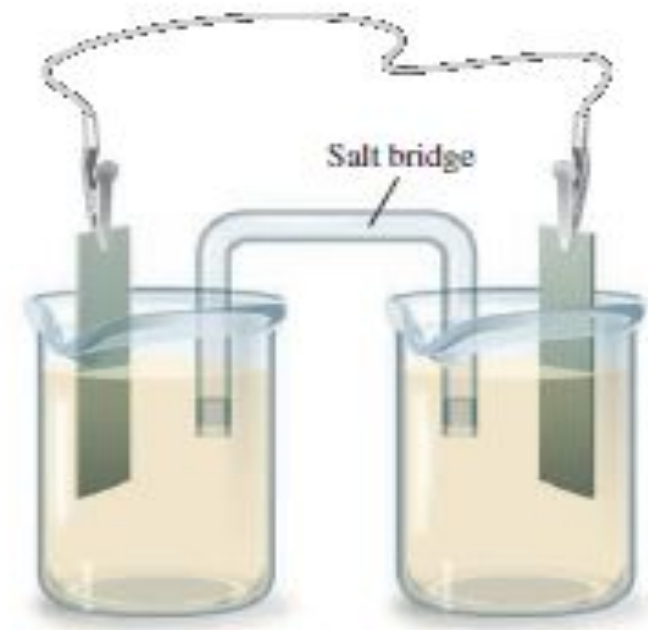


The current stops flowing because of charge buildups in the two compartments.

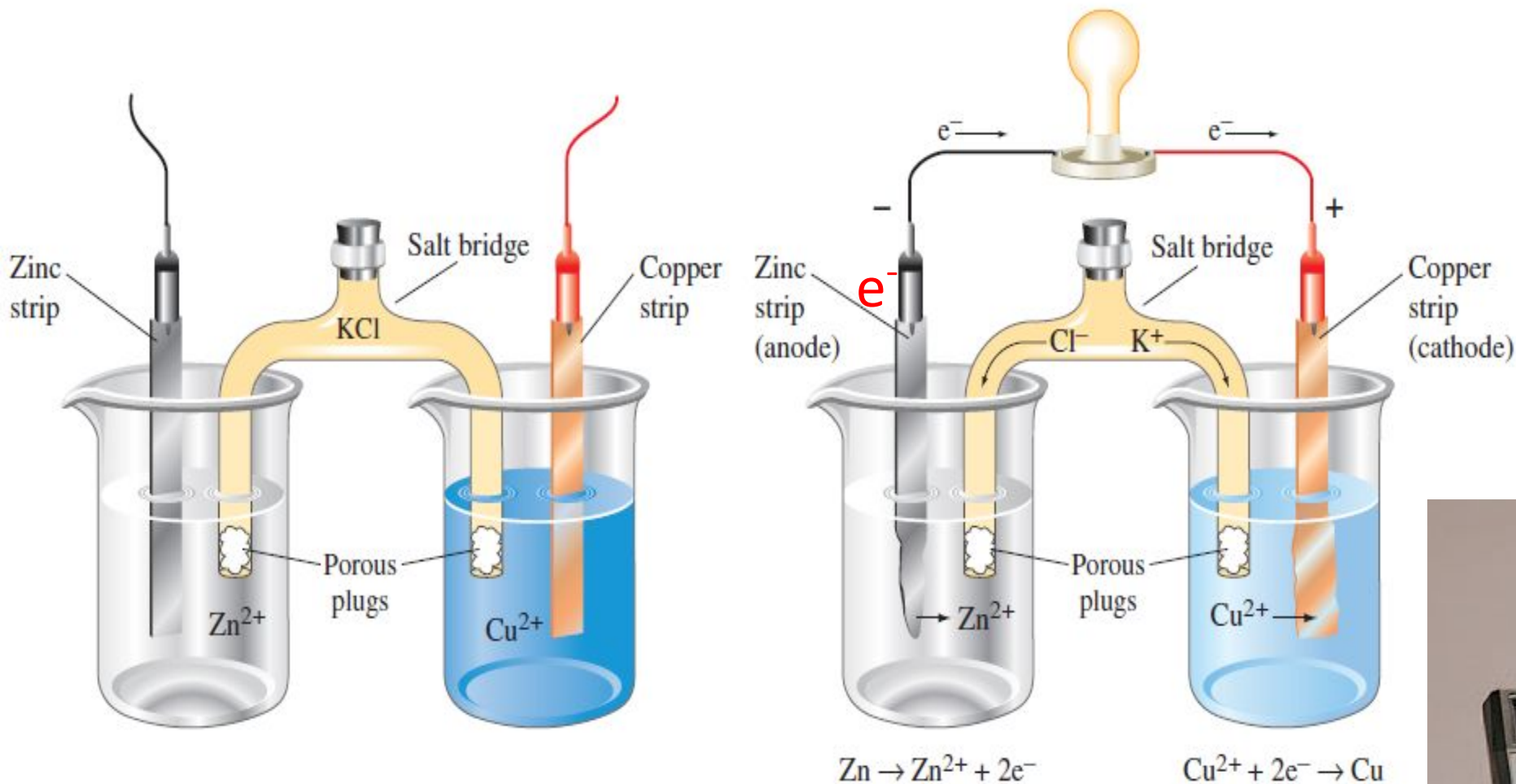


We can, however, solve this problem very simply. The solutions must be connected so that ions can flow to maintain the net charge of zero in each compartment.

This connection might involve a salt bridge (a U-tube filled with an electrolyte) or a porous disk in a tube connecting the two solutions



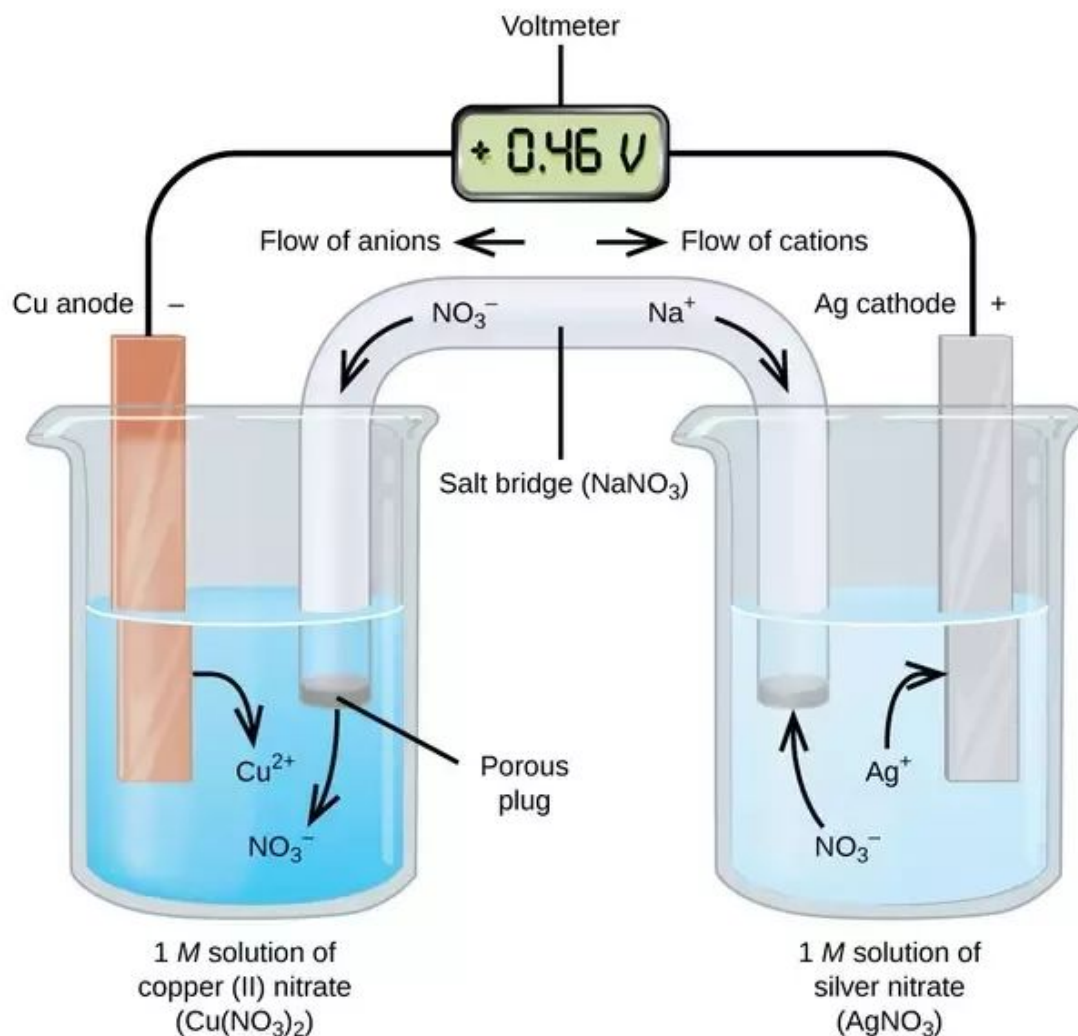
- The strongest oxidizing agent present in the cell always undergoes a reduction at the cathode.
- The strongest reducing agent present in the cell always undergoes an oxidation at the anode.



A zinc=copper voltaic cell (A) A zinc electrode and a copper electrode, without an external circuit; there is no cell reaction. (B) When the two electrodes are connected by an external circuit (with a lightbulb), chemical reaction occurs. (C) A similar cell, with the bulb replaced by a voltmeter. Note the blue color of $\text{Cu}^{2+}(\text{aq})$.



A Theoretical Description of a Galvanic Cell

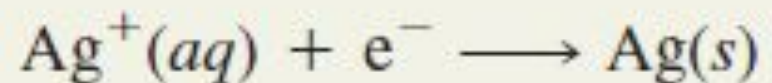


Evidence	Interpretation
The copper electrode decreases in mass and the intensity of the blue colour of the electrolyte increases.	Oxidation of copper metal is occurring: $\text{Cu}_{(s)} \rightarrow \text{Cu}_{(aq)}^{2+} + 2e^-$ blue
The silver electrode increases in mass as long, silver-coloured crystals grow.	Reduction of silver ions is occurring: $\text{Ag}_{(aq)}^+ + e^- \rightarrow \text{Ag}_{(s)}$
A blue colour slowly moves up the U-tube from the copper half-cell to the silver half-cell and the solution remains electrically neutral.	Copper(II) ions move toward the cathode. Negative ions (anions) move toward the anode.
A voltmeter indicates that the silver electrode is the cathode (positive) and the copper electrode is the anode (negative).	Electrons move from the copper electrode to the silver electrode.
An ammeter shows that the electric current flows between the copper electrode and the silver electrode.	Electrons leave the copper half-cell and enter the silver half-cell.

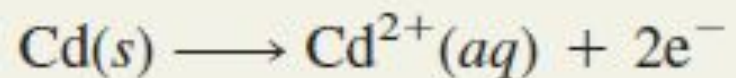
Sketching and Labeling a Voltaic Cell

A voltaic cell is constructed from a half-cell in which a cadmium rod dips into a solution of cadmium nitrate, $\text{Cd}(\text{NO}_3)_2$, and another half-cell in which a silver rod dips into a solution of silver nitrate, AgNO_3 . The two half-cells are connected by a salt bridge. Silver ion is reduced during operation of the voltaic cell. Draw a sketch of the cell. Label the anode and cathode, showing the corresponding half-reactions at these electrodes. Indicate the electron flow in the external circuit, the signs of the electrodes, and the direction of cation migration in the half-cells.

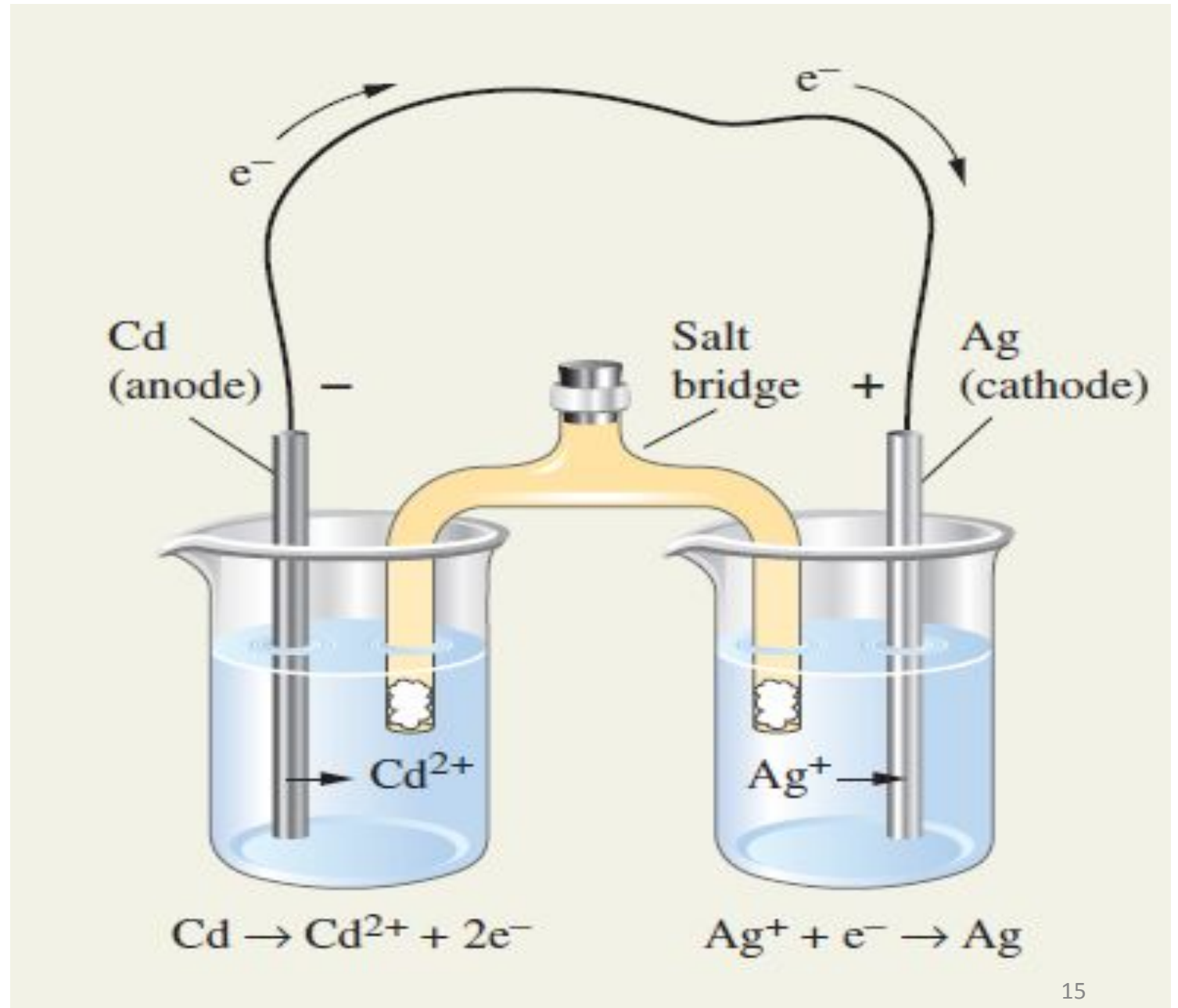
Because silver ion is reduced at the silver electrode, the silver electrode is the cathode. The half-reaction is



The cadmium electrode must be the anode (electrode at which oxidation occurs); the half-reaction is

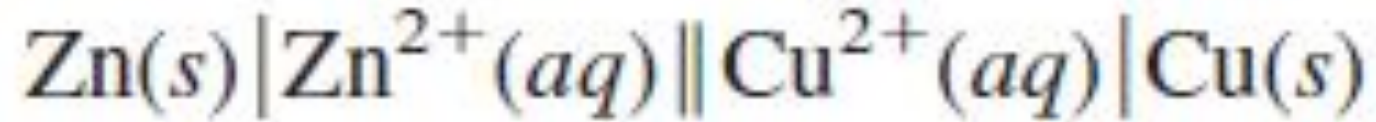


Now that you have labeled the electrodes, you can see that the electron flow in the external circuit is from the cadmium electrode (anode) to the silver electrode (cathode). Positive ions will flow in the solution portion of the circuit opposite to the direction of the electrons. **The sketch for this cell, including the labeling, is given in Figure**

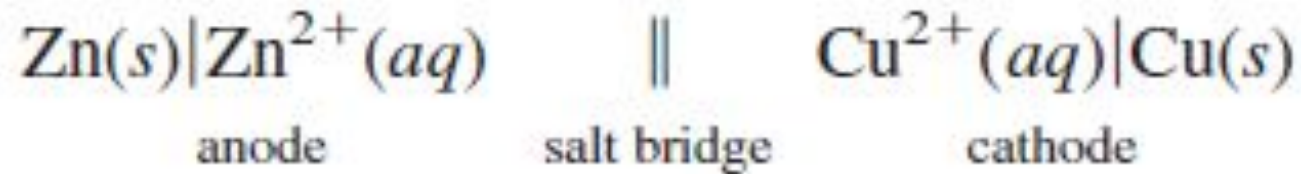


Notation for Voltaic Cells

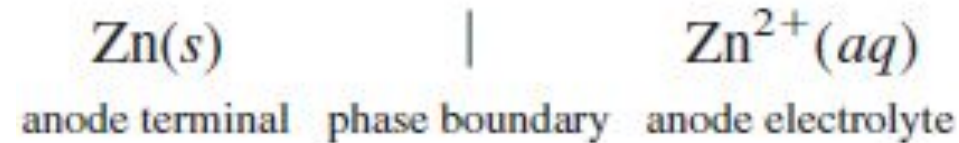
It is convenient to have a shorthand way of designating particular voltaic cells. The cell described earlier, consisting of a zinc metal–zinc ion half-cell and a copper metal–copper ion half-cell, is written



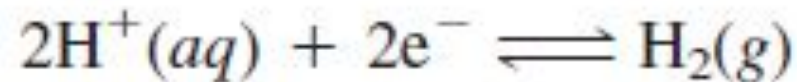
In this notation, the anode, or oxidation half-cell, is always written on the left; the cathode, or reduction half-cell, is written on the right. The two electrodes are electrically connected by means of a salt bridge, denoted by two vertical bars.



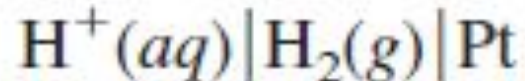
The cell terminals are at the extreme ends in this cell notation, and a single vertical bar indicates a phase boundary—say, between a solid terminal and the electrode solution. For the anode of the same cell, you have



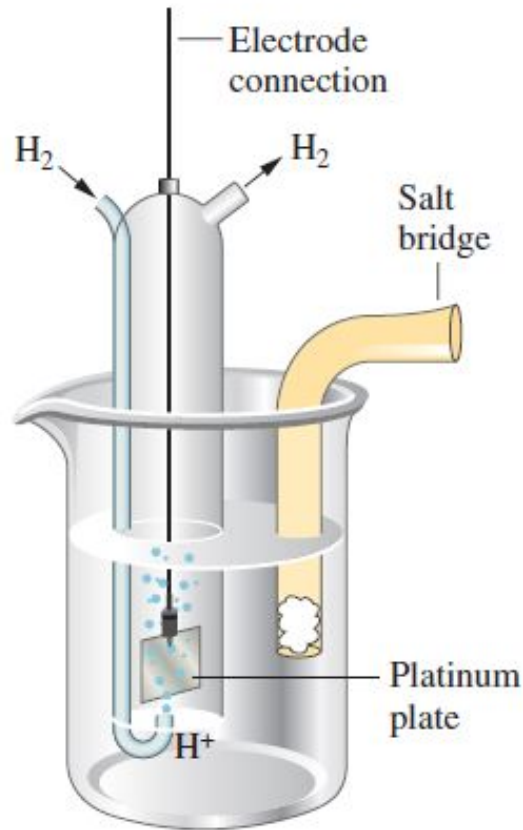
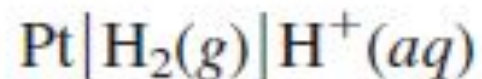
When the half-reaction involves a gas, an inert material such as platinum serves as a terminal and as an electrode surface on which the half-reaction occurs. The platinum catalyzes the half-reaction but otherwise is not involved in it. Figure 20.5 shows a hydrogen electrode; hydrogen bubbles over a platinum plate that is immersed in an acidic solution. The cathode half-reaction is



The notation for the hydrogen electrode, written as a cathode, is

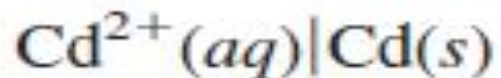
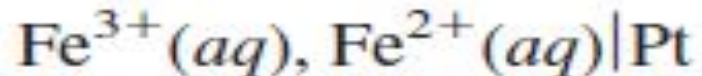
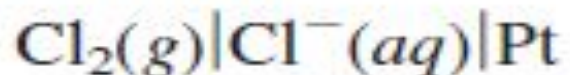


To write such an electrode as an anode, you simply reverse the notation:

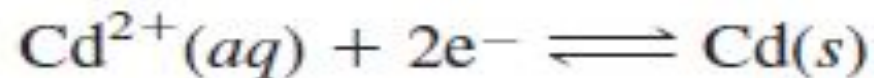
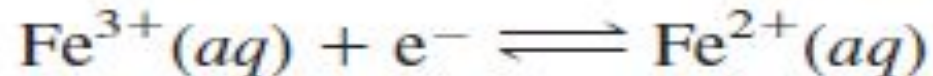
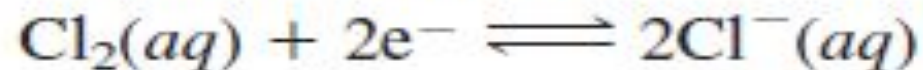


Here are several additional examples of this notation for electrodes (written as cathodes). A comma separates ions present in the same solution. We will write the cathode with oxidized species before reduced species, in the same order as in the half-reaction.

Cathode

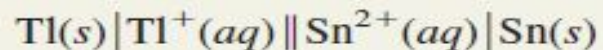


Cathode Reaction

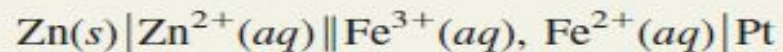


Writing the Cell Reaction from the Cell Notation

a. Write the cell reaction for the voltaic cell

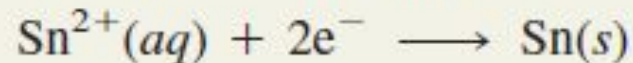
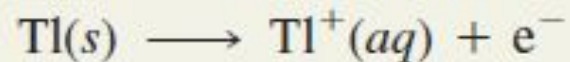


b. Write the cell reaction for the voltaic cell



The cell notation gives the species involved in each half-reaction. Complete and balance each half-reaction, then multiply by factors so that when you add the half reactions, the electrons cancel. The result is the cell reaction.

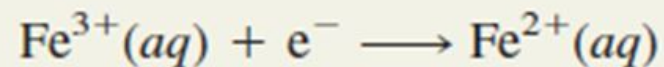
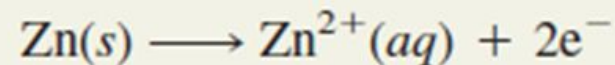
a. The half-cell reactions are



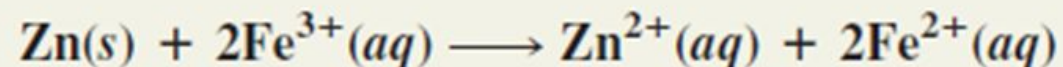
Multiplying the anode reaction by 2 and then summing the half-cell reactions gives



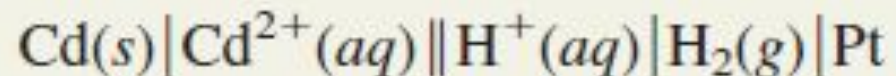
b. The half-cell reactions are



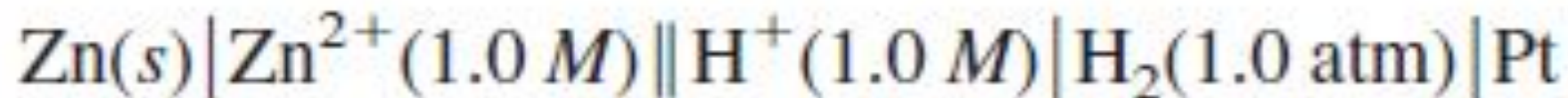
and the cell reaction is



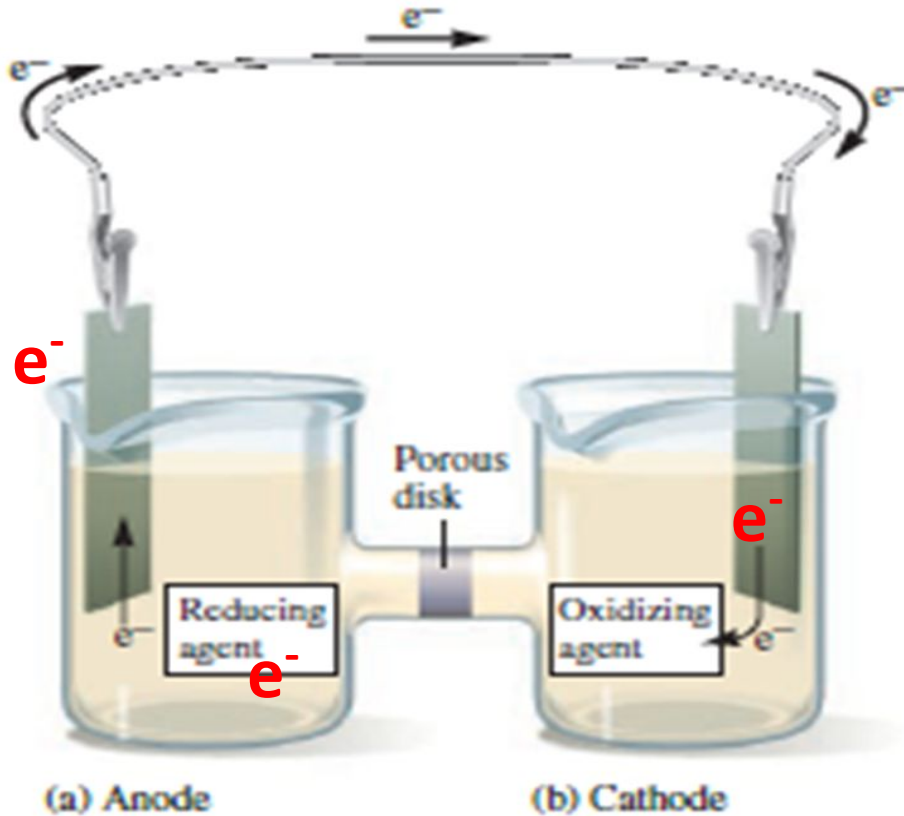
Give the overall cell reaction for the voltaic cell



To fully specify a voltaic cell, it is necessary to give the concentrations of solutions or ions and the pressure of gases. In the cell notation, these are written within parentheses for each species. For example,



Cell Potential / Electromotive Force



In galvanic cell, oxidizing agent in one compartment pulls electrons through a wire from a reducing agent in the other compartment.

The “pull,” or driving force, on the electrons is called the cell potential (E_{cell}), or the electromotive force (emf), of the cell.

The unit of electrical potential is the volt (abbreviated V), which is defined as 1 joule of work per coulomb of charge transferred.

$$E_{\text{cell}} = E_{\text{red (cathode)}} + E_{\text{ox (anode)}}$$

$$E_{\text{cell}} = E_{\text{red (cathode)}} - E_{\text{red (anode)}}$$

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

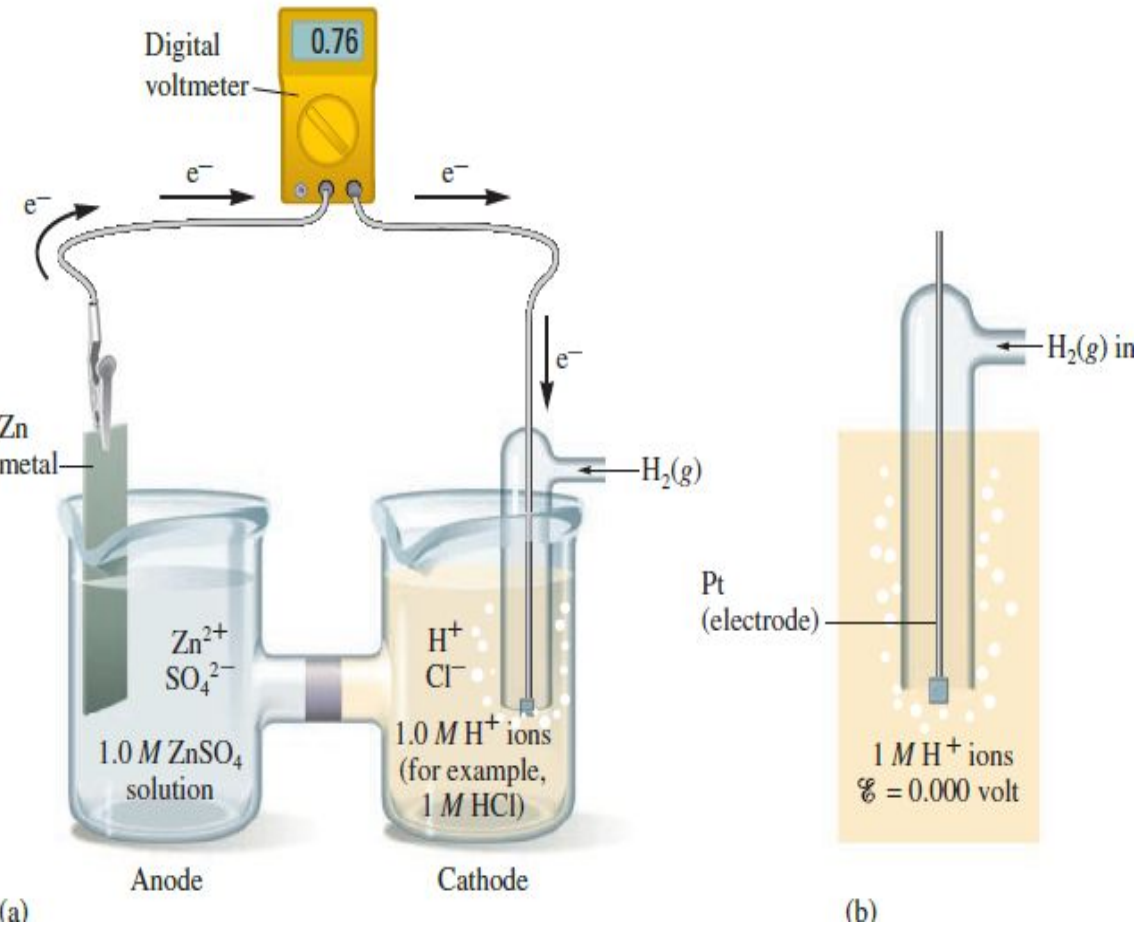
Standard Electrode Potentials

The standard emf, E°_{cell} , is the emf of a voltaic cell operating under standard-state conditions:

- **solute concentrations are each 1 M**
- **gas pressures are each 1 atm,**
- **and the temperature has a specified value—usually 25°C**

Note the superscript degree sign ($^\circ$), which signifies standard-state conditions.

How to measure standard electrode potential



Standard hydrogen electrode (SHE) is the most commonly used reference electrode since its potential is exactly equal to Zero at all temperatures.

Since the potential of SHE is zero, it forms the perfect basis to calculate cell potentials using different electrodes or different concentrations.



$$E_{\text{cell}}^{\circ} = E_{\text{H}^+ \rightarrow \text{H}_2}^{\circ} + E_{\text{Zn} \rightarrow \text{Zn}^{2+}}^{\circ}$$

$\uparrow$ $\uparrow$ $\uparrow$
 0.76 V 0 V 0.76 V

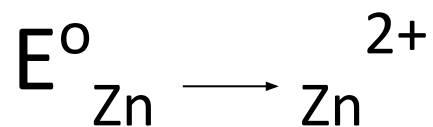
Standard Reduction Potentials at 25°C (298 K) for Many Common Half-reactions

Half-reaction	$\mathcal{E}^\circ$ (V)	Half-reaction	$\mathcal{E}^\circ$ (V)
$\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}^-$	2.87	$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$	0.40
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$	1.99	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	0.34
$\text{Co}^{3+} + \text{e}^- \rightarrow \text{Co}^{2+}$	1.82	$\text{Hg}_2\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Hg} + 2\text{Cl}^-$	0.27
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$	1.78	$\text{AgCl} + \text{e}^- \rightarrow \text{Ag} + \text{Cl}^-$	0.22
$\text{Ce}^{4+} + \text{e}^- \rightarrow \text{Ce}^{3+}$	1.70	$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{SO}_3 + \text{H}_2\text{O}$	0.20
$\text{PbO}_2 + 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{e}^- \rightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$	1.69	$\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$	0.16
$\text{MnO}_4^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{MnO}_2 + 2\text{H}_2\text{O}$	1.68	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.00
$\text{IO}_4^- + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{IO}_3^- + \text{H}_2\text{O}$	1.60	$\text{Fe}^{3+} + 3\text{e}^- \rightarrow \text{Fe}$	-0.036
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51	$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$	-0.13
$\text{Au}^{3+} + 3\text{e}^- \rightarrow \text{Au}$	1.50	$\text{Sn}^{2+} + 2\text{e}^- \rightarrow \text{Sn}$	-0.14
$\text{PbO}_2 + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{Pb}^{2+} + 2\text{H}_2\text{O}$	1.46	$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$	-0.23
$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	1.36	$\text{PbSO}_4 + 2\text{e}^- \rightarrow \text{Pb} + \text{SO}_4^{2-}$	-0.35
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.33	$\text{Cd}^{2+} + 2\text{e}^- \rightarrow \text{Cd}$	-0.40
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	1.23	$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	-0.44
$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$	1.21	$\text{Cr}^{3+} + \text{e}^- \rightarrow \text{Cr}^{2+}$	-0.50
$\text{IO}_3^- + 6\text{H}^+ + 5\text{e}^- \rightarrow \frac{1}{2}\text{I}_2 + 3\text{H}_2\text{O}$	1.20	$\text{Cr}^{3+} + 3\text{e}^- \rightarrow \text{Cr}$	-0.73
$\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$	1.09	$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.76
$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightarrow \text{VO}^{2+} + \text{H}_2\text{O}$	1.00	$2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$	-0.83
$\text{AuCl}_4^- + 3\text{e}^- \rightarrow \text{Au} + 4\text{Cl}^-$	0.99	$\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$	-1.18
$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{NO} + 2\text{H}_2\text{O}$	0.96	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	-1.66
$\text{ClO}_2 + \text{e}^- \rightarrow \text{ClO}_2^-$	0.954	$\text{H}_2 + 2\text{e}^- \rightarrow 2\text{H}^-$	-2.23
$2\text{Hg}^{2+} + 2\text{e}^- \rightarrow \text{Hg}_2^{2+}$	0.91	$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$	-2.37
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$	0.80	$\text{La}^{3+} + 3\text{e}^- \rightarrow \text{La}$	-2.37
$\text{Hg}_2^{2+} + 2\text{e}^- \rightarrow 2\text{Hg}$	0.80	$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$	-2.71
$\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$	0.77	$\text{Ca}^{2+} + 2\text{e}^- \rightarrow \text{Ca}$	-2.76
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}_2$	0.68	$\text{Ba}^{2+} + 2\text{e}^- \rightarrow \text{Ba}$	-2.90
$\text{MnO}_4^- + \text{e}^- \rightarrow \text{MnO}_4^{2-}$	0.56	$\text{K}^+ + \text{e}^- \rightarrow \text{K}$	-2.92
$\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$	0.54	$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}$	-3.05
$\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$	0.52		

Representing redox potentials

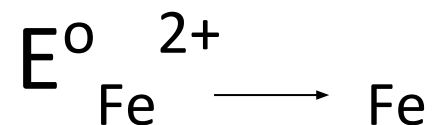
Standard Oxidation potential

$$E^{\circ}_{\text{cell Zn}/\text{Zn}^{2+}}$$



Standard Reduction potential

$$E^{\circ}_{\text{cell Fe}^{2+}/\text{Fe}}$$



Two important rule

Rule 1: When a half-reaction is reversed, the sign of E° is reversed.

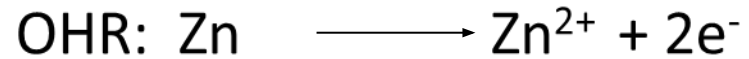
Ex: If $E^{\circ}_{\text{cell Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$ then, $E^{\circ}_{\text{cell Zn}/\text{Zn}^{2+}} = +0.76 \text{ V}$

Rule 2: When a half-reaction is multiplied by an integer, E° remains the same.

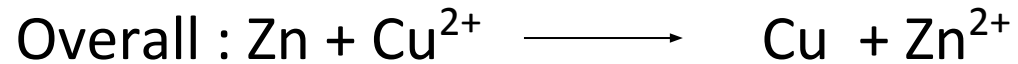
What is the potential for the following cell?



$$E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = 0.33\text{V}$$



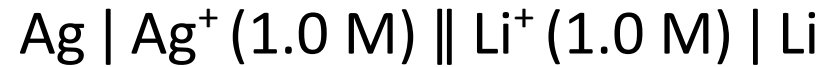
$$E_{\text{Zn}/\text{Zn}^{2+}}^{\circ} = 0.76 \text{ V}$$



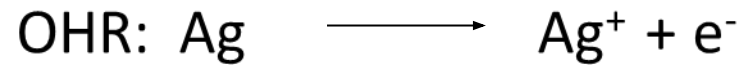
$$E_{\text{Cell}}^{\circ} = 1.09 \text{ V}$$

The positive potential confirms your observation that zinc metal reacts with cupric ions in solution to produce copper metal.

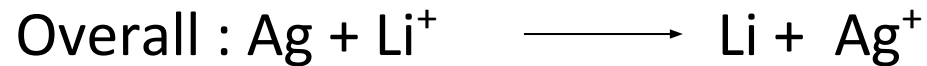
What is the potential for the following cell?



$$E_{\text{Li}^+/\text{Li}}^{\circ} = -3.05 \text{ V}$$



$$E_{\text{Ag}/\text{Ag}^+}^{\circ} = -0.80 \text{ V}$$



$$E_{\text{Cell}}^{\circ} = -3.85 \text{ V}$$

The negative potential indicates that this cell is not feasible. However, reverse reaction should be spontaneous.

A galvanic cell consists of a Mg electrode in a 1.0 M $\text{Mg}(\text{NO}_3)_2$ solution and a Ag electrode in a 1.0 M AgNO_3 solution. Calculate the standard emf of this cell at 25°C

Strategy: At first it may not be clear how to assign the electrodes in the galvanic cell. From Table 18.1 we write the standard reduction potentials of Ag and Mg and apply the diagonal rule to determine which is the anode and which is the cathode.

Solution: Standard reduction potential are

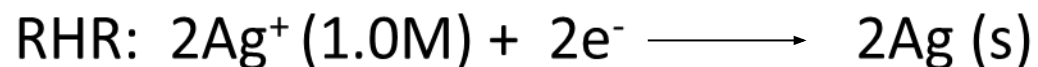


$$E_{\text{Ag}^+/\text{Ag}}^{\circ} = 0.80 \text{ V}$$



$$E_{\text{Mg}^{2+}/\text{Mg}}^{\circ} = -2.37 \text{ V}$$

As the reduction potential of Ag is higher than Mg, reduction of Ag will happen in cathode and Oxidation of Mg will happen in Anode.



$$E_{\text{Ag}^+/\text{Ag}}^{\circ} = +0.80 \text{ V}$$



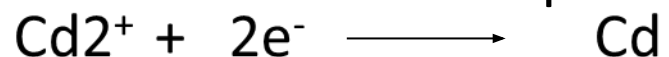
$$E_{\text{Mg}/\text{Mg}^{2+}}^{\circ} = -(-2.37 \text{ V})$$



The positive value of E° shows that the forward reaction is favored

- What is the standard emf of a galvanic cell made of a Cd electrode in a 1.0 M $\text{Cd}(\text{NO}_3)_2$ solution and a Cr electrode in a 1.0 M $\text{Cr}(\text{NO}_3)_3$ solution at 25°C?

Solution: Standard reduction potential are

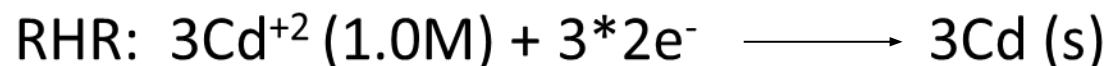


$$E_{\text{Cd}^{2+}/\text{Cd}}^{\circ} = -0.40 \text{ V}$$



$$E_{\text{Cr}^{3+}/\text{Cr}}^{\circ} = -0.73 \text{ V}$$

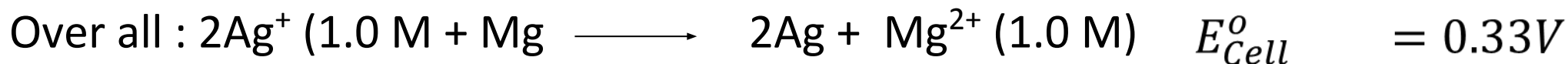
As the reduction potential of Ag is higher than Mg, reduction of Ag will happen in cathode and Oxidation of Mg will happen in Anode.



$$E_{\text{Cd}^{2+}/\text{Cd}}^{\circ} = -0.40 \text{ V}$$



$$E_{\text{Cr}^{3+}/\text{Cr}}^{\circ} = 0.73 \text{ V}$$



The positive value of E° shows that the forward reaction is favored

Practice Problems:

- What is the standard emf of a galvanic cell made of a Cd electrode in a 1.0 M $\text{Cd}(\text{NO}_3)_2$ solution and a Cr electrode in a 1.0 M $\text{Cr}(\text{NO}_3)_3$ solution at 25°C?
- Calculate the standard emf of a cell that uses the Mg/Mg^{2+} and Cu/Cu^{2+} half-cell reactions at 25°C. Write the equation for the cell reaction that occurs under standard-state conditions
- Calculate the standard emf of a cell that uses Ag/Ag^+ and Al/Al^{3+} half-cell reactions. Write the cell reaction that occurs under standard-state conditions.
- Using the data from standard reduction potential table, predict whether 1 M HNO_3 will dissolve gold metal to form a 1 M Au^{3+} solution.

Consider a galvanic cell based on the reaction

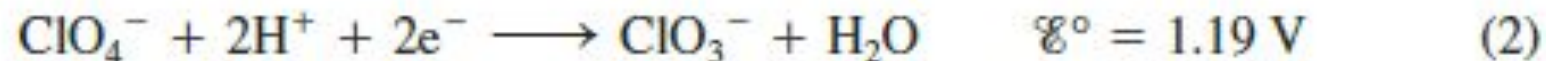
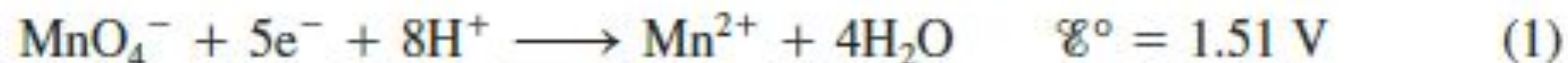


Give the balanced cell reaction, and calculate $\mathcal{E}^\circ$ for the cell. Standard reduction potential of Al and Mg are -1.66 and -2.37, respectively.

b. A galvanic cell is based on the reaction



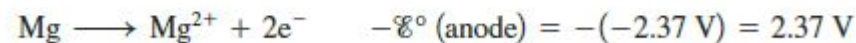
The half-reactions are



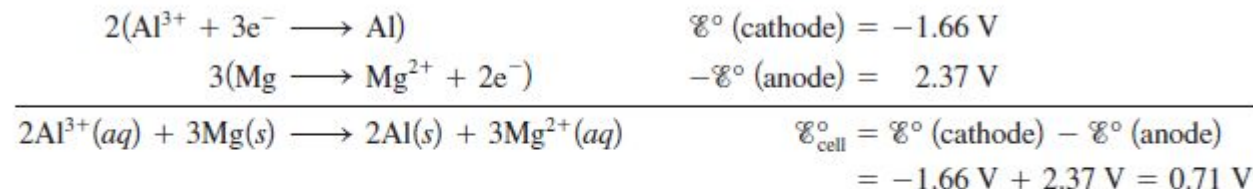
Give the balanced cell reaction, and calculate $\mathcal{E}^\circ$ for the cell.

Solution

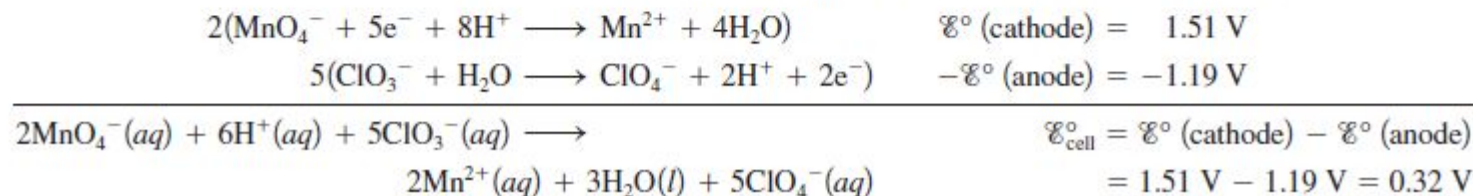
- a. The half-reaction involving magnesium must be reversed and since this is the oxidation process, it is the anode:



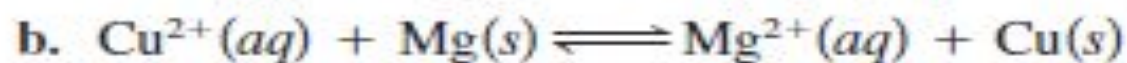
Also, since the two half-reactions involve different numbers of electrons, they must be multiplied by integers as follows:



- b. Half-reaction (2) must be reversed (it is the anode), and both half-reactions must be multiplied by integers to make the number of electrons equal:



37. Sketch the galvanic cells based on the following overall reactions. Show the direction of electron flow, and identify the cathode and anode. Give the overall balanced equation. Assume that all concentrations are 1.0 M and that all partial pressures are 1.0 atm.



38. Sketch the galvanic cells based on the following overall reactions. Show the direction of electron flow, the direction of ion migration through the salt bridge, and identify the cathode and anode. Give the overall balanced equation. Assume that all concentrations are 1.0 M and that all partial pressures are 1.0 atm.

